Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The port 53 is unreachable**.** This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: UDP port 53 unreachable length 254. The port noted in the error message is used for DNS service. The most likely issue is that no service is listening on the receiving DNS port.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
The incident was reported at approximately 1:24 p.m. Customers reported that they were unable to access the website yummyrecipesforme.com and received a “destination port unreachable” error message. The IT team used tcpdump to capture and analyze network traffic. They examined UDP and ICMP packets to identify the source of the connectivity problem and determine why DNS requests were failing. The network analysis showed that UDP traffic to port 53 was unreachable. Port 53 is used by DNS services. The ICMP responses indicated that the DNS server was not accepting requests on that port. A possible cause is a DNS server outage, a firewall misconfiguration blocking UDP port 53, or a denial-of-service (DoS) attack that made the DNS service unavailable.